

VATTIKUTI MOHIT

AI/ML Engineer

📍 Kakinada, Andhra Pradesh ✉ mohit.v2112@gmail.com ☎ +91-6303934715 in LinkedIn 📄 GitHub

📁 Portfolio 🏆 HackerRank

Summary

Final-year CSE (Artificial Intelligence) student targeting AI/ML Engineer roles, with hands-on experience building machine learning and deep learning systems end-to-end – from custom transformer pretraining and multi-GPU (FSDP) training to a batched, low-latency inference-serving pipeline. Also comfortable working close to the metal, having constructed a full TCP/IP stack from raw bytes in C++, which reinforces systems fundamentals that support efficient ML infrastructure and MLOps.

Education

Pragati Engineering College, Andhra Pradesh *B.Tech in CSE – Artificial Intelligence* Sept 2023 – Present
GPA: 7.2/10

Technical Skills

Languages: Python, C++, JavaScript, SQL, C

AI/ML: Machine Learning, Deep Learning, PyTorch, TensorFlow, Scikit-learn, Hugging Face, Computer Vision (Grad-CAM, MediaPipe, Librosa)

Systems & Infra: Distributed Training (FSDP, torchrun), MLOps, FastAPI/REST APIs, TCP/IP, ICMP, TUN/TAP, CMake, Git/GitHub

Web: React, Node.js, HTML/CSS, Three.js

Projects

LLM-From-Scratch: Pretraining, Distributed Training & Serving

- Implemented a decoder-only transformer from scratch (RoPE, RMSNorm, SwiGLU, GQA) with a custom BPE tokenizer and verified multi-process FSDP pretraining via torchrun.
- Engineered a FastAPI batched-inference server with KV-caching, achieving a 2.36x inference speedup and 2.25x serving throughput over unbatched generation, confirmed via exact-match correctness tests against the unbatched baseline.
- Diagnosed and fixed a DPO reward-hacking failure (100% preference accuracy masking a degenerate, collapsed model) via reward-trajectory analysis and multi-seed ablations.
- Tech Stack:** PyTorch, FastAPI, torchrun/FSDP, custom BPE, DPO

📄 Code · ▶ Live Demo

Userspace TCP/IP Stack (tinystack)

- Developed a complete TCP/IP stack in C++ running entirely in userspace via macOS utun, with zero OS networking calls and zero external libraries – raw bytes only.
- Implemented the full RFC 793 TCP state machine (LISTEN → SYN_RECEIVED → ESTABLISHED → FIN → CLOSED), reaching sub-millisecond ICMP round-trips (0.21 ms average, 0% loss over 40 packets).
- Designed an internal packet router supporting both a TCP server (HTTP/1.0, port 8080) and an outbound TCP client, enabling full HTTP request/response cycles outside the kernel network stack.
- Tech Stack:** C++20, CMake, ICMP, TCP, RFC 793, TUN/TAP, non-blocking I/O, RFC 1071 checksum

📄 Code · ▶ Terminal Demo

Audio Deepfake Detector

- Trained a CNN-based spoofed-speech classifier on the ASVspoof 2019 LA dataset (121,000+ samples), reaching a 23.93% Equal Error Rate on attack types withheld from training.
- Architected a Mel-spectrogram pipeline (Librosa) feeding a 4-block CNN with Batch Normalization and Global Average Pooling, cutting parameter count by ~90% while correcting a 9:1 class imbalance via weighted sampling.
- Fixed a data-leakage bug via systematic audio trimming, added Grad-CAM explainability, and deployed an interactive Streamlit demo on Hugging Face Spaces.
- Tech Stack:** Python, PyTorch, Librosa, NumPy, Pandas, Scikit-learn, Streamlit

📄 Code · ▶ Live Demo

Achievements

- Earned a 5-star HackerRank rating across 30+ SQL challenges, improving query execution time by 40% through indexing and query rewrites; optimized a core DSA algorithm from $O(n^2)$ to $O(n \log n)$, reducing runtime by 30% on benchmark inputs.
- Pretrained 3 transformer variants (13.7M / 29.3M / 91.1M params) on a real 82MB code corpus (Tesla T4 GPU, Kaggle free tier) under a matched 32.77M-token budget, showing a clean, above-noise scaling law: validation loss dropped monotonically (3.4199 → 3.0166 → 2.5673), with both deltas measuring 13–15x the noise floor.

Certifications & Coursework

Certifications: Python & Generative AI – Infosys SpringBoard • GitHub Foundations Certificate

Coursework (Cisco Networking Academy, CCNA track): Introduction to Networks; Switching, Routing & Wireless Essentials; Enterprise Networking, Security & Automation